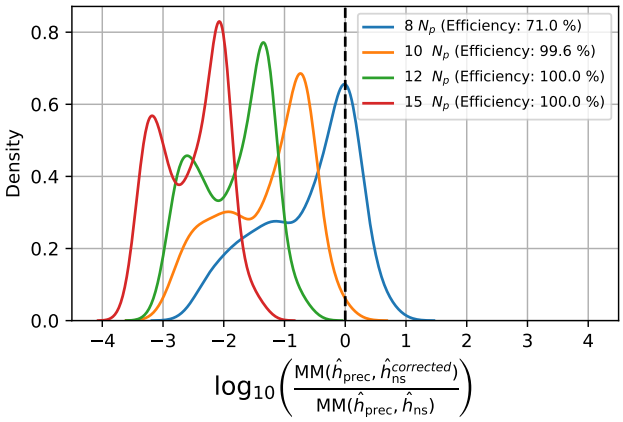

FIG. 10. We demonstrate the efficiency of the cubic spline method based on the number of nodal points (N_p) used to correct the baseline model. The x-axis represents the logarithm of the ratio of mismatches: the mismatch after correction compared to the mismatch before correction. The function $\mathrm{MM}(h_1, h_2)$ is defined in equation A2. Here, h_{prec} represents the injected waveform strain with precessing signals, while h_{ns} refers to the strain generated with the same parameters except the spin parameters set to zero. $h_{\mathrm{ns}}^{\mathrm{corrected}}$ indicates the corrected strain using waveform-error parametrization with N_p frequency nodal points. A logarithmic mismatch ratio less than one indicates an improvement, meaning that the corrected model is closer to the injected waveform h_{prec}. The vertical dashed line serves as a reference. The efficiency is defined as the percentage of simulation points where an improvement is achieved (i.e., where the logarithmic mismatch ratio is less than 0).

eccentricity [116–119]. The most general description of a BBH merger can include other environmental effects such as matter accretion, another binary or heavy object nearby, or a merger near a supermassive BH near the host galaxy's center.

However, the waveform model used in PE might not fully describe the reality. We want to see if the WF-Error parametrization can capture the deviation from the actual model because the missing effects are not considered in the waveform model. In order to test this scenario, we injected a set of precessing signals with components coming from the uniform distribution $s_{1,2}^{x,y,z} \in \mathcal{U}(-0.1, 0.1)$. We use the `IMRPhenomPv2` model and other parameters described in the previous section. We call the injected model h_{prec}. We expect bias in the measured parameters for a precessing injection if we use a recovery model that does not include spins in the waveform description: $h_{\mathrm{no-spins}}$, where we set spin parameters to zero in the IMRPhenomPv2 waveform model. For the PE analysis, we use the priors described in table III. For the precessing PE runs, we use isotropic spin priors for all the spin components $s_{1,2}^{x,y,z}$. Our selection of spin parameters aims to modify the reference waveform model, which includes the missing physics. Our goal is not to model deviations based on an actual astrophysical distribution, but rather to investigate whether we can incorporate such deviations for individual signals, if they are present.

In order to test if cubic-spline corrections to the $h_{\mathrm{no-spin}}$ work, we simulate 500 realizations of the GW strain h_{prec}. We then estimate the amplitude and phase differences $(\delta\tilde{A}_{\mathrm{rel}}(f), \delta\tilde{\phi}_{\mathrm{rel}}(f))$ with respect to the non-spinning waveform template $\hat{h}_{\mathrm{no-spin}}$. We then choose a different number of nodal points, N_p, and apply the correction to the non-spinning template, using cubic splines generated using these nodal points. We define the mismatch between two normalized templates $\hat{h}_1$ and $\hat{h}_1$ as,

$$\mathrm{MM}(\hat{h}_1, \hat{h}_1) = 1 - \langle \hat{h}_1 | \hat{h}_2 \rangle, \tag{22}$$

where $\hat{h}$ is a normalized template such that the inner product $\langle \hat{h} | \hat{h} \rangle = 1$. We estimate the mismatch between the waveform strain before and after the correction, i.e., $\mathrm{MM}_1 = \mathrm{MM}(\hat{h}_{\mathrm{prec}}, \hat{h}_{\mathrm{no-spin}})$, and $\mathrm{MM}_2 = \mathrm{MM}(\hat{h}_{\mathrm{prec}}, \hat{h}_{\mathrm{no-spin}}^{\mathrm{corrected}})$. When the cubic spline corrections to $h_{\mathrm{no-spin}}$ bring the model closer to the injected signal h_{prec}, the ratio $\frac{\mathrm{MM}_2}{\mathrm{MM}_1}$ is less than unity, or the logarithm of this ratio is less than 0. In Figure 10, we show the distribution of the ratio $\frac{\mathrm{MM}_2}{\mathrm{MM}_1}$ for different number of nodal points used for cubic spline curves. For $N_p = 8$, we get this ratio less than one for $> 70\%$ simulated precessing signals. For $N_p = 10$, more than 99% of cases show the improvement. This number was 100% for $N_p \geq 12$. This indicate that, in principle, using ≥ 10 nodal points to apply corrections to baseline non-spinning waveform model should be enough to incorporate missing physics considered in this specific study.

We now focus on PE runs done in this study with ten realizations of precessing signals generated from the distribution described earlier in this subsection. Figure 11 shows that when we use a non-spinning waveform model $h_{\mathrm{no-spins}}$ to recover precessing injections, we get bias in the recovered chirp mass parameter $\mathcal{M}^{\mathrm{z}}$ in detector frame. However, when we use WF-Error parametrizations along with $h_{\mathrm{no-spins}}$ waveform model, we notice that, for rel-phase parametrization, it results in broader marginalized posterior samples. However, abs-phase parametrization cannot make their posterior samples broad enough to correct the biases for this specific example. We observe that for injections with a significant mismatch ($\geq 1\%$) to the reference model h_{ref}, the correction remains inadequate to account for the missing physics fully. We will revisit this issue later in this subsection.

We assess the performance of the WF-Error parametrization using the ratio of systematic errors to statistical errors, expressed as $|\Delta\mathcal{M}^{\mathrm{z}}|/\sigma_{\mathcal{M}^{\mathrm{z}}}$, with two runs consist of one that employs the $h_{\mathrm{no-spin}}$ waveform model and another that applies the same model with the rel-phase parametrization. As demonstrated in equation (6), systematic errors become important when this ratio is of the order of unity. In Figure 12, on the x-axis, we show the mismatch between two templates: h_{prec} and $h_{\mathrm{no-spins}}$, keeping the values of non-spinning parameters

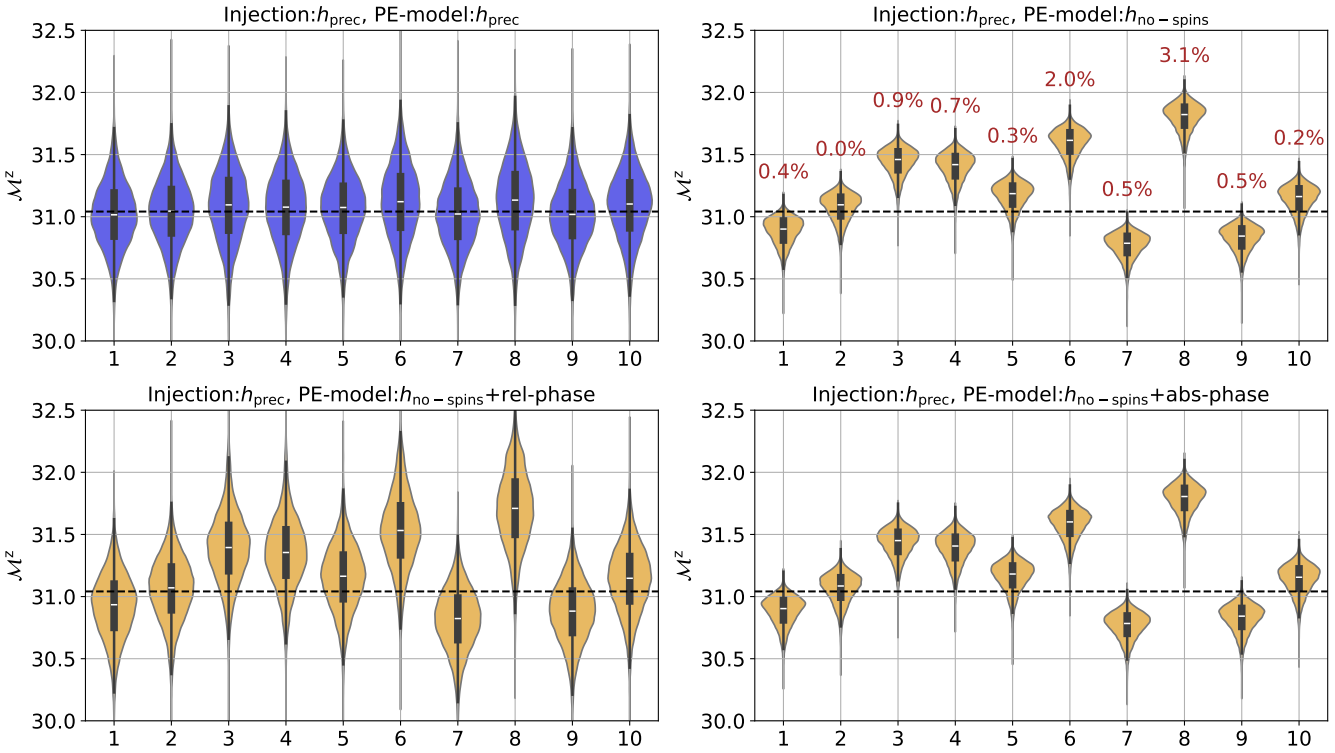

FIG. 11. We use ten precessing injections in the HLV network. The top left panel shows the posterior samples from each PE run where we use the precessing recovery model: $h_{\rm prec}$. We use the nonspinning model $h_{\rm no-spins}$ for PE in the top right panel. The bottom panels use a nonspinning model along with the WF-Error framework. The bottom left plot shows recovery with rel-phase parametrization, and the bottom right model uses abs-phase parametrization. The horizontal dashed line represents the injected chirp mass ($\mathcal{M}^z$) value in the detector frame. The x-axes represent the injection simulation IDs. We use all the parameters listed in Table III for these PE runs. We also use component spin parameters $s_{1,2}^{x,y,z}$ for the precessing recovery. We use the same priors for $(\delta\tilde{A}_i, \delta\tilde{\phi}_i)$ parameters for both the WF-Error parametrizations in bottom panels. The numbers above the data points in the top right panel represent mismatch (%) between the injected template waveform $h_{\rm prec}$ and the corresponding non-spinning template $h_{\rm no-spin}$.

the same. It quantifies how much overlap there is between the waveform templates. On the y-axis, we show the quantity $|\Delta\mathcal{M}^z|/\sigma_{\mathcal{M}^z}$ (left panel), and the bias $\mathcal{M}^z$ (right panel). The bias is estimated as the difference between the injected value and the median of the 1D marginalized posterior samples. The ratio $|\Delta\mathcal{M}^z|/\sigma_{\mathcal{M}^z}$ increases when there is a larger mismatch between the two templates. It shows a significant drop in the ratio when the rel-phase WF-Error parametrization is used with the $h_{\rm no-spins}$ waveform model. This drop is a combination of both the effects: broadening of posterior samples (larger σ), as well as a decrease in bias $\Delta\mathcal{M}^z$ (See right panel of Figure 12). In this set of PE runs, we do not find significant biases in any other parameter except $\mathcal{M}^z$.

The functional form of the parameters $(\delta\tilde{A}, \delta\tilde{\phi})$ for the simulation set considered in this section is shown in the appendix B. In our PE runs, though we notice consistent decrease in the bias value $|\Delta\mathcal{M}^z|$ and the ratio $|\Delta\mathcal{M}^z|/\sigma_{\mathcal{M}^z}$, we focus on two injections where the original mismatch was more than 1%. The current framework fails to fully mitigate the systematic bias (ideally $|\Delta\mathcal{M}^z| \to 0$). Upon further investigation, we find that this could be because of a combination of i) A most generalized waveform modification should be applied on both polarizations $(h^+, h^\times)$ independently, and ii) a large prior ranges for $\delta\tilde{A}_i, \delta\tilde{\phi}_i$ are needed to incorporate significant deviations for large mismatch. We leave the details of such an investigation for a follow-up study.

We want to emphasize that this is an extreme example of the proof of concept. However, it indicates that we can switch on appropriate WF-Error parametrization in PE analysis whenever we are unsure if the waveform model captures the realistic description. If the 'true' model differs from the reference model in such a way that the WF-Error parametrizations can capture it, we should be able to account for potential biases in the source parameters of the binary system.

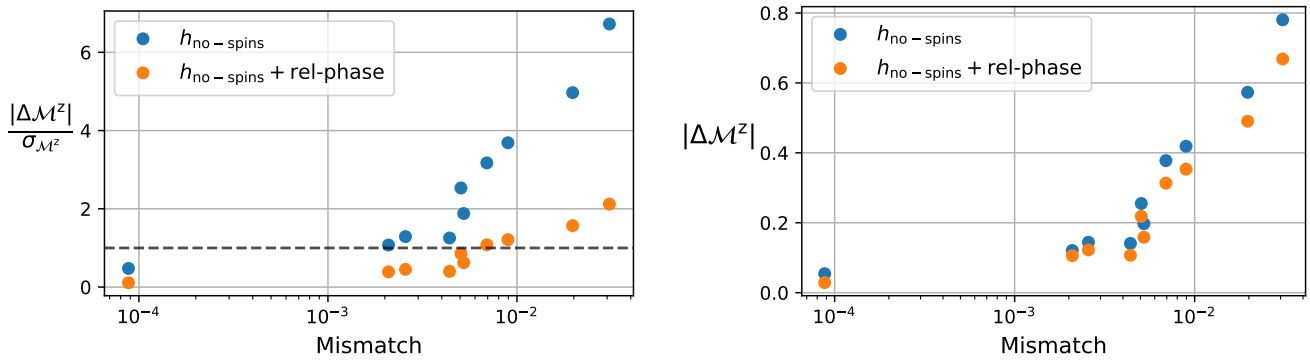

FIG. 12. For the injections and recovery shown in Figure 11, on the x-axis, we show the mismatch between the waveform generated with non-spinning and precessing parameter values. We kept all other parameters the same (see text). **Left panel**: The y-axis shows the ratio of the absolute value of the bias ($\Delta\mathcal{M}^z = \mathcal{M}^z_{median} - \mathcal{M}^z_{inj}$) to the standard deviation of the 1D marginalized posterior samples of $\mathcal{M}^z$. Horizontal dashed line represent the the line where $\Delta\mathcal{M}^z = \sigma_{\mathcal{M}^z}$. Right panel: The y-axis shows the absolute value of the bias $\Delta\mathcal{M}^z$.

VI. COMPARISON WITH OTHER APPROACHES IN THE LITERATURE

This section compares our framework with existing approaches in the literature regarding waveform systematics. We categorize the types of systematic errors that waveform modeling may encounter, focusing on at least two primary categories:

- **Type-A errors**: These errors are realized when GW waveform morphology of a true signal h_{true} can not be achieved by the waveform model $h_{\text{model}}(\mathbf{\Theta})$. Mathematically,

$$\left\langle \hat{h}_{\text{true}} | \hat{h}(\mathbf{\Theta}) \right\rangle \quad < \quad 1 \;\; \forall \;\; \mathbf{\Theta}, \tag{23}$$

  where $\hat{h}$ is the normalized strain such that $\langle \hat{h} | \hat{h} \rangle = 1$.

- **Type-B errors**: The waveform model can generate the true signal's morphology, but for the wrong parameters $\mathbf{\Theta}$. These errors are challenging to address, particularly if we rely on only one waveform model or if all available waveform models exhibit similar systematic errors.

Current approaches to mitigate the waveform systematics can be categorized in several ways, with some methods fitting into multiple categories. However, we will list them under their primary category:

- **Account for differences between waveform models**: In some regions of parameter space, such as high spins or highly unequal mass ratios, waveform models fall short of the accuracy requirements [120], and different waveform models may give inconsistent results in PE [6]. Several approaches try to account for these differences, e.g., by combining the posterior samples with appropriate weighting [6, 33, 34] or using hyperparameters [35–40] to sample across different waveform models. It is important to understand that if all waveform models share common systematics, particularly type-A errors, then these methods will not be able to address those systematics. However, if one of the waveform models exhibits type-B errors, this approach can help mitigate the impact of waveform systematics by diluting the contribution from that model.

- **Numerical relativity calibration errors**: Other approaches include using Gaussian process regression techniques to model deviation from the reference waveform [41]. The deviations can be modeled by interpolating from differences found with accurate templates, e.g., numerical relativity simulations, or by modeling the numerical relativity calibration parameters used in waveform models based on effective one-body (EOB) approaches [42, 43] These approaches may fail to mitigate type-A errors if the NR simulations used for the waveform model calibration themselves contain the systematics e.g. thermal systematics in binary neutron star waveforms [121, 122].

- **Probabilistic waveform models**: Another suggested approach is that waveform developers provide the probabilistic models across the parameter space. For each waveform call, a bunch of curves with different weights associated with the model or an error band as a function of time or frequency can be provided [44].

Our framework provides a generic, data-driven approach in which the cubic splines can account for various systematic errors, including waveform systematics.